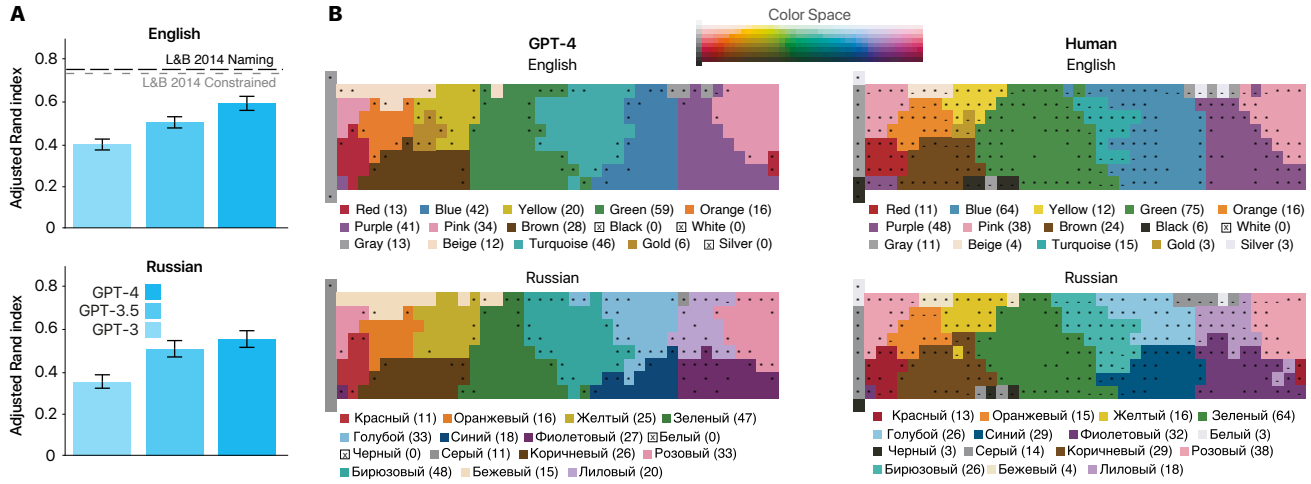

Fig. 3. Color naming experiment using 330 Munsell colors from the World Color Survey (top, color space). **A.** Adjusted Rand index illustrating the alignment between human and LLM experiments (95% CIs). The dashed lines for English represent lab-based free naming and forced-choice naming experiments collected by Lindsey and Brown (31) (data reproduced with permission). **B.** Data comparison between humans and LLMs in Russian and English. Participants and LLMs were shown colors and were asked to choose from the same 15-color list. The count of chosen colors for each option is given in parentheses. The color of a response cluster in the maps represents its average color.

future research directions. First, while in our work we experimented with three different types of behavioral data (similarity, explanations, and naming), there are other perceptual measures that one could consider (for example, odd-one-out triplet judgments (36)). Future work could explore these measures and interrogate to what extent they too yield a consistent representation. Second, our work is restricted to population-level averages as a leading-order analysis, a follow-up study could look into the natural variability of the LLM judgments and see to what extent they capture individual-level differences in humans. Third, while our work provides some evidence for LLMs' ability to capture cross-cultural differences, it remains to be seen how far this holds for other languages, especially those that are underrepresented (37). Finally, it is important to point out that the very same massive training pipelines that make LLMs a powerful proxy of human language also make them particularly susceptible to inheriting biases (38). Researchers should be particularly cautious when interpreting the patterns of behavior elicited from LLMs and should always benchmark them against genuine human data.

To conclude, our work showcases how LLMs can be used to explore the limits of what information about the world can be recovered from language alone, and more broadly, it highlights the potential of combining human and machine experiments in order to understand fundamental questions in cognitive science.

Materials and Methods

GPT prompt elicitation. The general structure of the prompt elicitation template for the similarity experiments was: one sentence describing the dataset (e.g., *"people described pairs of colors using their hex code."*), one sentence describing the similarity rating task and scale (e.g., *"how similar are the colors in each pair on a scale of 0-1 where 0 is completely dissimilar and 1 is completely similar?"*), three sets of three lines each corresponding to two stimuli and their actual similarity rating taken from behavioral experiments which serve as few-shot in-context examples, and an additional set of three lines corresponding to the pair of target stimuli and an associated empty rating field for the model to fill in (*"Color one: #FF0000. Color two: #A020F0. Rating:"*). These were necessary to ensure that the model provided numerical values, and in all cases consisted of three fixed and randomly chosen comparisons so that most of the content was left for the model to produce. For each pair of target stimuli, we elicited ten ratings from each GPT model. Across all repetitions of all pairs of stimuli for a given dataset, we used only the same three in-context examples to ensure that the model is exposed to only a very small fraction of the similarity judgments against which its ratings were compared (see SI for full prompts).

For the color-naming experiments, we first elicited 15 basic color names from GPT-4 using the prompt *"Name 15 basic colors."* and a temperature of 0 (to get the highest probability answers). We then had GPT-4 name the hex code corresponding to each of the WCS colors using the following prompt: *"Here is a list of 15 basic color names: <shuffled basic color list>. Which of these names best describes the following color: <hex-code>? Respond only using the name."* We repeated this prompt ten times for each WCS color with the basic color list shuffled each time and temperature set to the default 0.7 to elicit ten names per WCS color. We repeated the full procedure with both prompts translated to Russian (GPT-4 also responded to both of these in Russian; see SI for full prompt).

Stimuli. The six human similarity datasets we considered come in two flavors – direct (dis-)similarity ratings and confusion matrices – and from two sources – previous psychological studies from the literature and newly collected datasets. Confusion matrices provide an alternative way to compute similarity scores between stimuli by counting the number of times a stimulus x is confused for a stimulus y. By normalizing the counts one gets confusion probabilities p_{xy} which can be converted into similarity scores using the formula $s_{xy} = \sqrt{p_{xy}p_{yx}/p_{xx}p_{yy}}$ (39, 40).

Colors This dataset was taken from (25) (also reproduced in (24)) and comprised direct similarity judgments across a set of 14 colors with wavelengths in the range $434 - 674$ nanometers. We converted wavelengths into RGB using the script at https://hasanyavuz.ozderya.net/?p=211 and then we used the `webcolors` Python package to convert into hex codes. To get better coverage of the color wheel in Figure 2B we extended the space to 23 color stimuli by interpolating between the original colors in the dataset and eliciting an extended similarity matrix from GPT-4.

Pitch This dataset was collected and made publicly available very recently by a subset of the authors in (41) (see details below). It contains similarity judgments over pairs of 25 harmonic complex tones (10 partials and 3dB/octave roll-off) over a two octave range from C4 (60 MIDI; 261.626 Hz) to C6 (84 MIDI; 1046.502 Hz). The

pitch values were separated by 1 semitone steps to account for the fact that pitch perception is logarithmic (33) where the mapping between frequencies f in Hertz and pitch p in semitones are given by $p = 12 \log_2 f/440 + 69$.

Vocal consonants This dataset was also collected via an online study (see below) and comprised similarity judgments over 16 recordings of vocal consonants taken from the International Phonetic Association[‡]. The vocal consonants considered were b (**b**ay), p (**p**ay), m (**m**ay), n (**n**o), g (**g**o), k (ca**k**e), d (**d**ie), t (**t**ie), f (**f**ee), v (**v**ow), s (**s**o), θ (**th**igh), ð (**th**ey), ʒ (**J**acques), and ʃ (**sh**ow). The recordings came from two speakers, one male and one female.

Loudness We accessed this dataset via (40) which itself takes the data from (26). The dataset comes in the form of a confusion matrix over 8 pure tones of different loudness values ranging from 71.1 to 74.6 decibels.

Taste This dataset was also accessed via (40) and is taken from (28). The data comes in the form of a confusion matrix over 10 flavors described to participants as salt, salt-substitute, MSG, quinine, acid, sugar, artificial sweetener, salt-sugar, acid-sugar, and quinine-sugar.

Timbre This dataset was assembled in (27) based on 1217 subject's judgments from 5 prior publications. It comprises dissimilarity judgments over 12 instrument timbres: clarinet, saxophone, trumpet, cello, French horn, oboe, flute, English horn, bassoon, trombone, violin, and piano.

Behavioral experiments. To collect similarity judgments over pitch and vocal consonants we deployed two online experiments on Amazon Mechanical Turk (AMT). Overall, 55 participants completed the pitch study and 64 participants completed the vocal consonants study. In addition, we collected color naming data in Russian and British English participants using Prolific[§]. Overall, we recruited 154 participants of which 103 were UK participants and 51 were Russian participants. Experiments were implemented using PsyNet[¶] and Dallinger[‖]. See additional details in SI.

Code and Data availability. All data and code used in this work can be accessed via the following link: https://tinyurl.com/fudaby5p, with the exception of the (31) color naming dataset as it is only available upon request from the authors. An interactive visualization of two-dimensional MDS spaces for the 6 modalities is available at: https://computational-audition.github.io/LLM-psychophysics/all-modalities.html. The human color naming data can be interactively explored via: https://computational-audition.github.io/LLM-psychophysics/color.html.

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[‡] https://www.internationalphoneticassociation.org/

[§] https://www.prolific.co

[¶] https://psynet.dev/

[‖] https://dallinger.readthedocs.io/

Extended Methods

Explicit GPT prompts. GPT prompt elicitation experiments were conducted using the OpenAI Text Completion (for GPT-3) and Chat Completion (for GPT-3.5 and GPT-4) APIs. The temperature was set to 0.7 for similarity judgments and 0 for color naming experiments. The exact prompt formats used are shown below.

Similarity judgments.

Color:

```
People described pairs of colors using their hex
codes.  How similar are the two colors in each
pair on a scale of 0-1 where 0 is completely
dissimilar and 1 is completely similar?  Respond
only with the numerical similarity rating.
Color 1:  #ff5700 Color 2:  #ff9b00 Rating 0.76
Color 1:  #b3ff00 Color 2:  #00ff61 Rating:  0.45
Color 1:  #FF0000 Color 2:  #00b2ff Rating:  0.02
Color 1:  <hex-code1> Color 2:  <hex-code2>
Rating:
```

Pitch:

```
People described pairs of musical notes using
their frequencies in hertz.
How similar are the musical notes in each pair on
a scale of 0-1 where 0 is completely dissimilar
and 1 is completely similar?

Note 1:  587.3295358348151 Hz
Note 2:  987.7666025122483 Hz
Rating:  0.46083740655517463

Note 1:  349.2282314330039 Hz
Note 2:  277.1826309768721 Hz
Rating:  0.743838237117938

Note 1:  415.3046975799451 Hz
Note 2:  987.7666025122483 Hz
Rating:  0.19874605585261726

Note 1:  <frequency1>
Note 2:  <frequency2>
Rating:
```

Vocal consonants:

```
People described vocal consonants using the
international phonetic alphabet (IPA).
How similar do the vocal consonants in each pair
sound on a scale of 0-1 where 0 is completely
dissimilar and 1 is completely similar?
Respond only with the numerical similarity
rating.

Vocal Consonant 1:  f
Vocal Consonant 2:  m
Rating:  0.5

Vocal Consonant 1:  n
Vocal Consonant 2:  ʒ
Rating:  0.40740740740740744

Vocal Consonant 1: ʃ
Vocal Consonant 2: ʃ
Rating:  1.0

Vocal Consonant 1:  <consonant1>
Vocal Consonant 2:  <consonant2>
Rating:
```

Loudness:

```
People described the loudness of pure tones in
decibels (dB).
How similar do the pure tones in each pair
sound on a scale of 0-1 where 0 is completely
dissimilar and 1 is completely similar?

Pure Tone 1:  72.6 dB
Pure Tone 2:  74.1 dB
Rating:  0.3495324720283043

Pure Tone 1:  74.6 dB
Pure Tone 2:  73.6 dB
Rating:  0.5055839477695901

Pure Tone 1:  74.1 dB
Pure Tone 2:  74.1 dB
Rating:  1.0

Pure Tone 1:  <loudness1>
Pure Tone 2:  <loudness2>
Rating:
```

Taste:

```
People described flavors they tasted using words.
How similar are the flavors in each pair on a
scale of 0-1 where 0 is completely dissimilar and
1 is completely similar?

Flavor 1:  quinine
Flavor 2:  artificial sweetener
Rating:  0.0

Flavor 1:  artificial sweetener
Flavor 2:  salt
Rating:  0.015433904145892428

Flavor 1:  quinine-sugar
Flavor 2:  acid-sugar
Rating:  0.2539115246067999

Flavor 1:  <flavor1>
Flavor 2:  <flavor2>
Rating:
```

Timbre:

```
People listened to pairs of musical instruments
and rated the similarity of their timbre.
How similar is the timbre of the instruments in
each pair on a scale of 0-1 where 0 is completely
dissimilar and 1 is completely similar?

Instrument 1:  Cello
Instrument 2:  Flute
Rating:  0.5604846433040316

Instrument 1:  Flute
Instrument 2:  Clarinet
Rating:  0.270932601836378

Instrument 1:  Trombone
Instrument 2:  Bassoon
Rating:  0.2893895067551666

Instrument 1:  <instrument1>
Instrument 2:  <instrument2>
Rating:
```